

Ultrahigh-speed all-optical wavelength conversion and format conversion for polarisation-shift-keying signal based on four-wave mixing in light-holding semiconductor optical amplifier

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Abstract: An ultrahigh-speed all-optical module, which can realise both the wavelength conversion of polarisation-shift-keying (PolSK) signal and the format conversion from non-return-to-zero (NRZ)-PolSK to return-to-zero (RZ)-PolSK, is proposed based on four-wave mixing (FWM) in light-holding semiconductor optical amplifier (SOA). This scheme is simple and free of pattern effect. The conversion efficiencies are improved by utilising a holding light. Using the polarisation-dependent broadband model, wavelength conversion and format conversion for PolSK signal are theoretically realised at 40 Gbit/s. Moreover, the effects of holding light, polarisation dependence of SOA and polarisation angle between input pump light and PolSK signal on the output performance of wavelength conversion and format conversion for PolSK signal are theoretically investigated in detail.

1 Introduction

Although amplitude-shift-keying (ASK) formats are simple and widely employed in optical systems and networks of all scales, they suffer the chromatic dispersion, non-linear effects and the noise in high-speed long-haul optical communication systems. In recent years, polarisation-shift-keying (PolSK) formats, which use the polarisation state of a fully polarised lightwave carrier as the information-bearing parameter, have been extensively investigated because of their unique virtues [1–5]. Extinction ratio and signal-to-noise ratio using PolSK formats are superior to using intensity formats. PolSK formats are less sensitive to phase noise of the transmitting lasers in comparison with other coherent transmission schemes. Using the PolSK signals, ultrafast multifunctional all-optical logic gates based on four-wave mixing (FWM) in SOAs are realised [6, 7]. So PolSK may become one of the novel chosen formats for future long-haul communications.

Ultrahigh-speed all-optical wavelength conversion is a key function in providing full scalability and flexible management in wavelength-division-multiplexing (WDM) networks. In addition, ultrahigh-speed all-optical format conversion, which can enhance the optical network flexibility, has received much interest recently. There have been a variety of wavelength conversion techniques and format conversion techniques using many approaches based on non-linear materials [8–14]. However, most wavelength conversion techniques are mainly focused on ASK formats and phase-shift-keying (PSK) formats. Also previous literature mostly

emphasise format conversion between ASK formats, PSK formats and frequency-shift-keying (FSK) formats. Up to now, only one wavelength conversion scheme for PolSK format based on FWM in a semiconductor optical amplifier (SOA) has been proposed [10]. Moreover to our knowledge no format conversion scheme between PolSK formats has been reported until now. So the studies of wavelength conversion and format conversion for PolSK formats are greatly expected.

It is difficult to preserve the original polarisation information of an input signal by using cross-gain modulation (XGM) and cross-phase modulation (XPM). FWM effect has the polarisation selectivity, as well as the merits of transparency to data rate and format. Therefore it can be used for wavelength conversion and format conversion for PolSK formats. Although high conversion efficiency has been demonstrated with SOAs of very high gain, the FWM effect in typical SOAs generally suffers from poor efficiency and noise figure. Significant improvement on the conversion efficiency and noise figure is required for practical applications.

In this paper, we propose an all-optical module which can realise both the wavelength conversion of PolSK signal and the format conversion from non-return-to-zero (NRZ)-PolSK to return-to-zero (RZ)-PolSK based on FWM in light-holding SOA. The wavelength conversion and the format conversion are implemented within the same module, which requires no additional adjustment to achieve compatibility. Such a multi-function module will be utmost cost-effective to future optical network nodes. Owing to the constant

intensity nature of the PolSK signal, pattern-dependent degradation can be reduced in this SOA-based device. Furthermore, this scheme can provide ultrafast operation in virtue of using the FWM effect. The proposed module only uses a single SOA, so it is simple and compact. A holding light can be utilised to extend the saturation power, and in turn allow obtaining high conversion efficiency in an SOA with moderate gain [15, 16]. The wavelength conversion and the format conversion for PolSK signal are theoretically realised at 40 Gbit/s using the polarisation-dependent broadband model. In addition, the effects of holding light, polarisation dependence of SOA and polarisation angle between input pump light and PolSK signal on the output performance of wavelength conversion and format conversion for PolSK signal are theoretically analysed.

2 Operation principle

Fig. 1 shows the schematic diagram of the proposed ultrahigh-speed all-optical wavelength conversion and format conversion for PolSK signal. A PolSK signal at λ_s and a pump light at λ_p , combined with a holding light at λ_a , are injected into the SOA. The FWM process occurs between λ_s and λ_p , and generates the conjugate and the satellite at $\lambda_c = 2\lambda_p - \lambda_s$ and $\lambda_{sat} = 2\lambda_s - \lambda_p$, respectively. The conjugate signal is filtered by a tunable filter (TF). It is well known that the FWM efficiency is strongly related to the polarisation angle between input optical waves, namely polarisation selectivity of FWM [17]. Two polarisation controls (PCs) are used to independently adjust the polarisations of input pump and PolSK signal. In this scheme, to obtain constant power for a converted PolSK signal, the polarisation angle between pump light and the logic '0' state of the input PolSK signal is set equal to that between the pump and '1' state of the signal. In this case, polarisation selectivity of FWM is satisfied in two polarisation directions of input PolSK signal, hence two FWM processes occur, respectively. If λ_s is a RZ-PolSK (or NRZ-PolSK) signal and λ_p is a synchronised clock pulse train (or a continuous-wave beam), the conjugate signal will preserve the polarisation information of the input PolSK signal. Thus the wavelength conversion for RZ-PolSK (or NRZ-PolSK) signal can be realised. If λ_s is a NRZ-PolSK signal and λ_p is a synchronised clock pulse train, the conjugate light will become a RZ-PolSK signal and the NRZ-PolSK to RZ-PolSK format conversion is realised. A holding light is used to improve the FWM efficiency. To prove the effectiveness of this scheme, a PC and a polarisation beam splitter (PBS) are used as the data decoder of PolSK format to demodulate the PolSK to ASK signals.

3 Simulation and results

In this scheme, the input signal is PolSK modulation, which includes two orthogonal polarisation states. So the

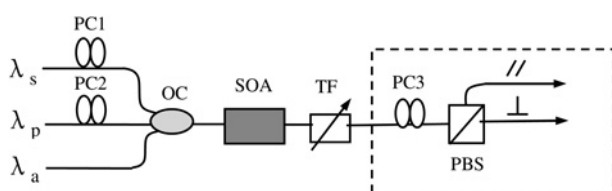


Fig. 1 Schematic diagram of the ultrahigh-speed all-optical wavelength conversion and format conversion for PolSK signal

polarisation dependence of the used SOA should be considered in this scheme. Consequently, the polarisation-dependent broadband model, which is presented in [7], is used in this paper. In this model, critical factors, for example, the polarisation dependence of the SOA, the material gain profile, the longitudinal variation of the optical field, the carrier density, photon density and the broadband spontaneous noise emission are considered. To simplify, we assume that the two polarisation components of the input PolSK signal are parallel to the principal axes of the SOA and correspond with TE mode and TM mode, respectively.

In our simulations, a 40 Gb/s 2^7-1 pseudorandom bit sequence (PRBS) NRZ-PolSK/RZ-PolSK signal and a synchronised 40 Gb/s OOK pump clock pulse train are considered. RZ-PolSK signal and OOK pump clock pulse train are both with a pulse duration equal to $T/4$, where T is the repetition period. The central wavelengths of the signal and pump are set to 1550 and 1551 nm, respectively. Thus the wavelengths of the conjugate and satellite signals are 1549 and 1552 nm, respectively. The wavelength of holding light is 1480 nm and its power is 50 mW. The injection current is 250 mA. The model confinement factors are set to 0.3 and 0.29 for TE mode and TM mode, respectively. In such a case, the SOA can be considered as polarisation-dependent since it has a polarisation sensitivity of 1.1164 dB. The other parameters are reported in [7].

Fig. 2 shows the simulation results of ultrahigh-speed all-optical wavelength conversion for RZ-PolSK signal with

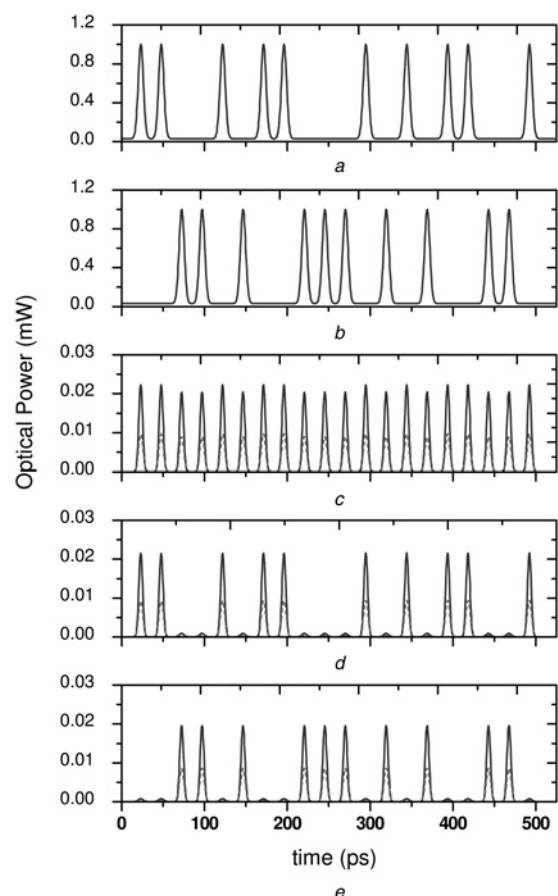


Fig. 2 Intensity waveforms of wavelength conversion

a, b Two polarisation components of input RZ-PolSK signal

c Converted RZ-PolSK signal

d, e Two constructive demodulation outputs of converted RZ-PolSK signal
With holding light: blue solid line, without holding light: red short dash line

(blue solid line) and without (red short dash line) the holding light. The peak powers of input PolSK signal and pump light are both set to 1 mW. Figs. 2a and b are the bit intensity sequences for two polarisation components of input RZ-PolSK signal. Its corresponding PolSK bit sequences are '11001011000101011001'. Fig. 2c is the converted RZ-PolSK signal. Figs. 2d and e are the constructive demodulation outputs of converted PolSK signal.

Fig. 3 shows the simulation results of ultrahigh-speed all-optical format conversion from NRZ-PolSK to RZ-PolSK with (blue solid line) and without (red short dash line) the holding light. The peak powers of input PolSK signal and pump light are set to 0.32 and 1 mW, respectively. Figs. 3a and b are the bit intensity sequences for two polarisation components of input NRZ-PolSK signal. Fig. 3c is the converted RZ-PolSK signal. Figs. 3d and e are the constructive demodulation outputs of converted RZ-PolSK signal.

From Figs. 2d, 2e and 3d, 3e, we can find that, although SOA is in deep saturation region, pattern effect nearly cannot be observed owing to the constant envelope of the PolSK signal. In addition, it is also observed that the output power of one polarisation component (see Figs. 2d and 3d) is larger than that of the other component (see Figs. 2e and 3e). This is because the polarisation dependence of SOA

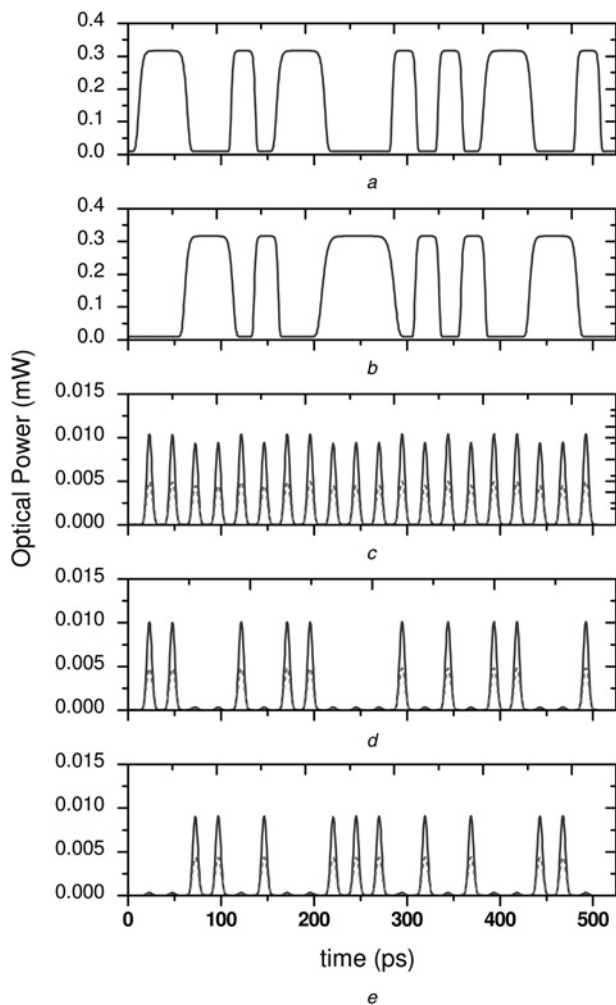


Fig. 3 Intensity waveforms of format conversion

a, b Two polarisation components of input NRZ-PolSK signal
c Converted RZ-PolSK signal
d, e Two constructive demodulation outputs of converted RZ-PolSK signal
With holding light: blue solid line, without holding light: red short dash line

leads to amplification difference between TE and TM. Consequently, although pattern effect can be ignored in this scheme, the amplitude of converted PolSK signals are not constant because of their including two polarisation components, as can be seen in Figs. 2c and 3c.

In addition, it can be seen clearly from Figs. 2c–e and 3c–e that the converted PolSK signals with the holding light are larger than those without the holding light. In other words, the conversion efficiencies are improved effectively by the holding light.

Although the residual polarisation dependence of the bulk SOA is small and can be almost avoided by designing the configuration of bulk SOA, in fact, it still exists to some extent. Therefore we further investigate the effect of polarisation dependence of SOA on the output performance of wavelength conversion and format conversion for PolSK signal. Figs. 4 and 5 show the wavelength conversion of RZ-PolSK signal and the format conversion from NRZ-PolSK to RZ-PolSK under $\Gamma^{\text{TE}} = 0.3$ and $\Gamma^{\text{TE}} = 0.27$, which corresponds to high polarisation-dependent case. Compared with the results in Figs. 2 and 3, it can be observed that, as the polarisation sensitivity of SOA increases, the difference of output powers between two polarisation components becomes larger; consequently, the wavelength conversion and the format conversion cannot be realised effectively.

An effective solution to reduce the effect of polarisation dependence of SOA in this scheme is making the polarisation angle between pump light and the logic '1' state of the input PolSK signal be properly larger than that between the pump and '0' state of the signal by adjusting PC2. Figs. 6 and 7 show that the simulation results of the

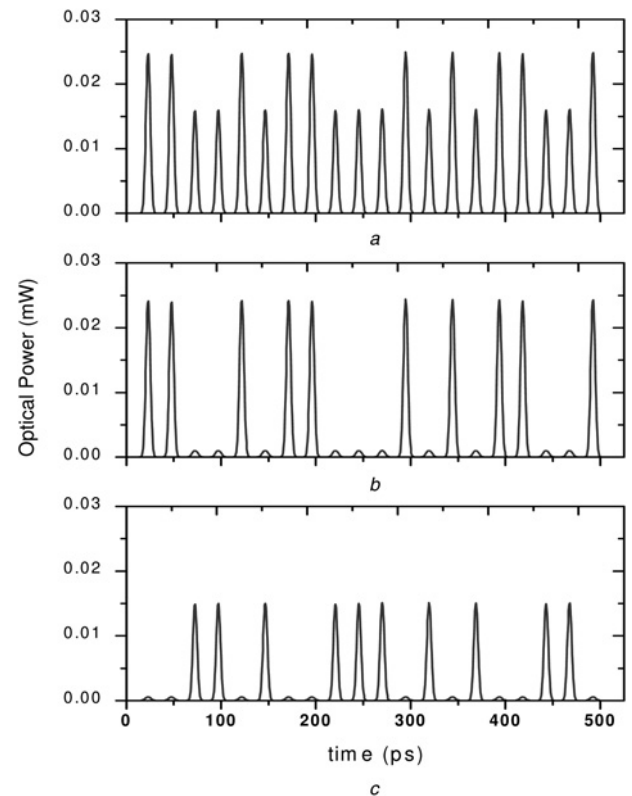


Fig. 4 Output intensity waveforms of wavelength conversion

a Converted RZ-PolSK signal
b and c Two constructive demodulation outputs of converted RZ-PolSK signal under $\Gamma^{\text{TE}} = 0.3$, $\Gamma^{\text{TM}} = 0.27$

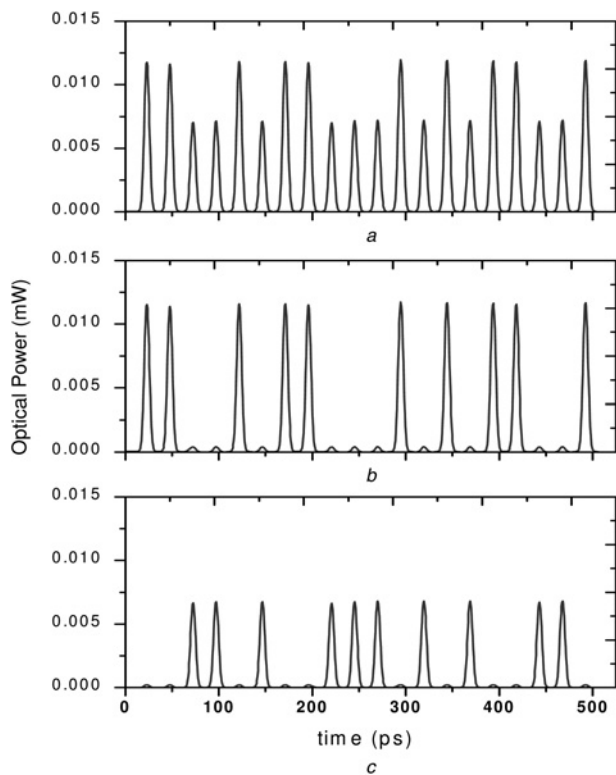


Fig. 5 Output intensity waveforms of format conversion

a Converted RZ-PolSK signal
b and *c* Two constructive demodulation outputs of converted RZ-PolSK signal under $\Gamma^{\text{TE}} = 0.3$, $\Gamma^{\text{TM}} = 0.27$

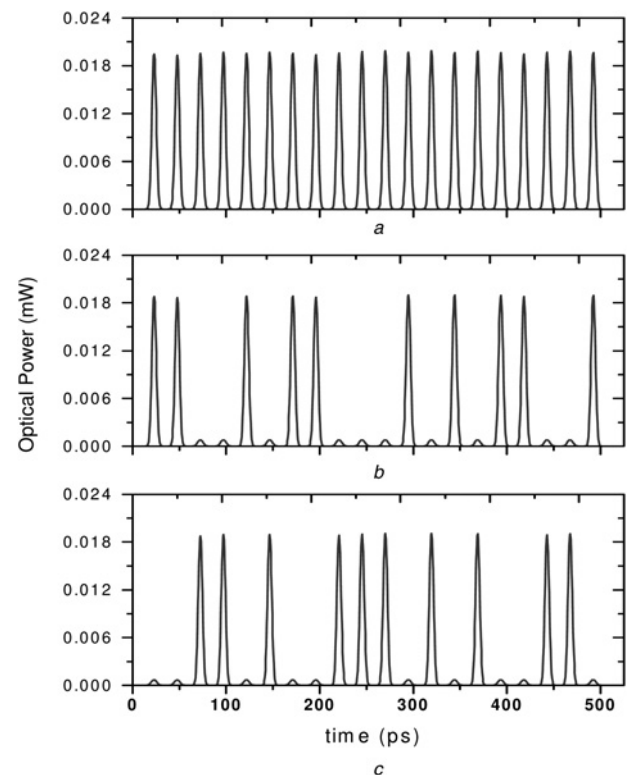


Fig. 7 Output intensity waveforms of format conversion

a Converted RZ-PolSK signal
b and *c* Two constructive demodulation outputs of converted RZ-PolSK signal under the polarisation angle between pump light and the logic '1' state of the input PolSK signal is 53°

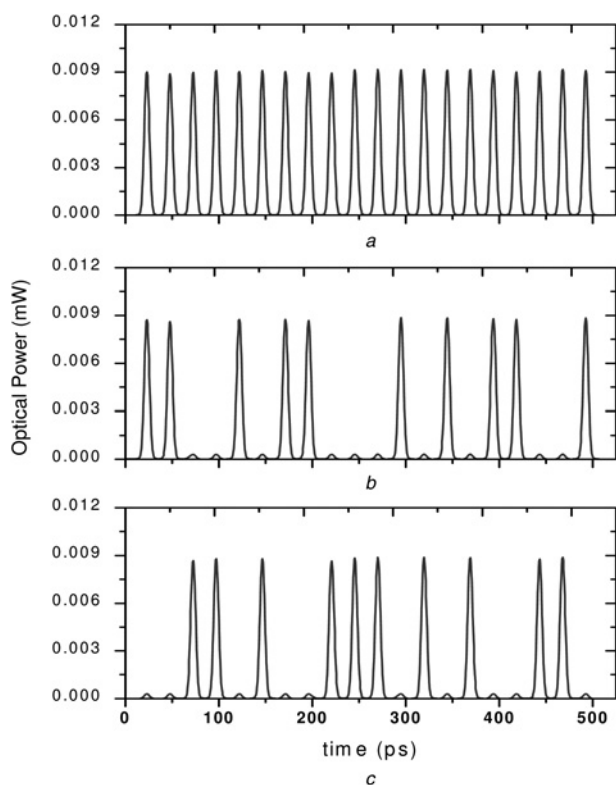


Fig. 6 Output intensity waveforms of wavelength conversion

a Converted RZ-PolSK signal
b and *c* Two constructive demodulation outputs of converted RZ-PolSK signal under the polarisation angle between pump light and the logic '1' state of the input PolSK signal is 52°

wavelength conversion and the format conversion under the polarisation angle between the pump light and the logic '1' state of the input PolSK signal are 52 and 53° , respectively. It is clearly seen that constant powers of converted PolSK signals can be achieved. This is because that, in this case, the component of pump (parallel to the logic '1' state of the input PolSK signal) is weaker than another component (parallel to the logic '0' state of the input PolSK signal). Therefore although the confinement factor of TE mode is larger than that of TM mode, the output converted PolSK signal powers of two polarisation components can be almost equal.

4 Conclusion

We have proposed a multifunction ultrahigh-speed all-optical module based on FWM in light-holding SOA. It has the ability to achieve both wavelength conversion for PolSK signal and format conversion from NRZ-PolSK to RZ-PolSK. Using the polarisation-dependent broadband model, we investigated the performance of the all-optical wavelength conversion and format conversion for PolSK signal at 40 Gbit/s. The results indicate that this scheme is free of pattern effect owing to using the PolSK modulation format. Since a holding light is injected into SOA, the conversion efficiencies of wavelength conversion and format conversion are improved effectively. Although this scheme is polarisation dependent, the effect of polarisation dependence of SOA can be reduced by adjusting two PCs. This scheme has the potential to be applied to future high bit rate optical networks.

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